

Planar Minimum Subdivisions for Normal-Surface Sectors

Linear One-Vertex Bounds, Quadratic Overlays,
Exact Enumeration, and Independent Certificates

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Abstract

The most recent minimum-envelope report in the ProveIt unknot-recognition program gives complete standard-ray enumeration when a supplied compatible quadrilateral sector has matching nullity at most two, and asks for the three-dimensional extension. We answer that specific question. After normalizing the sum of quadrilateral coordinates, a nullity-three sector has a planar projective section, possibly degenerate. Its non-link standard extreme rays are the vertices of a common subdivision by affine minima. If the polygon has b corners and the minimum diagram at vertex group v has f_v full-dimensional cells, we prove the sharp abstract bound

$$R \leq b + 2 \sum_v (f_v - 1) + \sum_{u < v} (f_u - 1)(f_v - 1).$$

The contracted normal-matching graph yields $R = O(k^2)$ for k allowed quadrilateral types, and $R \leq 7k + 6$ when only one vertex group varies, including one-vertex triangulations. An exact clipping-and-overlay producer and an independent minimum-tie-interval checker handle coincident forms, overlapping edges, and empty or lower-dimensional feasible sections. With preprojected lifting and deterministic sort-and-merge bookkeeping, a conservative post-kernel arithmetic bound is $O(1 + k^3 + (t + k)R)$ for t tetrahedra. The supplied set-based Python implementation has polynomial worst-case bit complexity; the sharper arithmetic claim does not assume a worst-case constant-time hash table. The delivered implementation matches all 9,807 eligible sectors in a declared 9,995-sector corpus, including 463 nullity-three cases; 607 independent replays and 328 certificate-mutation rejections supplement fresh Regina checks. Paired timings, their controls, and reproduction files accompany the article. The implemented overlay strengthens complete supplied-sector enumeration. An Euler-only precheck using the original quadrilateral corners admits post-kernel arithmetic cost $O(1 + t + k^2)$; this additional optimization is proved and proposed. Deducing a general quasi-polynomial unknot recognizer from this result still requires a controlled, complete sector-discovery or hierarchy theorem, which is not established here.

Research status. Proofs below establish the stated algebraic and geometric results. Sharpness is proved for rational potential systems; sharpness for realizable three-manifold triangulations is left open. The contribution is new relative to the audited repository and incoming minimum-envelope report. No claim of an exhaustive literature priority search is made.

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1 The result and the bottleneck it removes

The useful next step in the current repository is a complete algorithm for one more feasible dimension. The previous minimum-envelope package explicitly asks for an output-sensitive overlay algorithm in dimension three, including degenerate vertices, coincident faces, and intermediate-size bounds [8]. That question has a precise local meaning: fix a compatible choice of quadrilateral types, solve its matching equations, and enumerate *every* standard normal vertex ray supported there.

The maintained arrangement implementation introduces every pairwise equality between triangle potentials at the same global vertex. Most of those equalities can occur above the actual minimum. In a three-dimensional quadrilateral kernel, intersecting all pairs of these hyperplanes gives a quartic candidate bound in the sector size. Each surviving direction then requires canonical lifting and a standard-cone rank test. The method is complete, but it deliberately over-refines the relevant geometry.

We instead retain the actual minimum diagrams. A single diagram is a convex polygonal subdivision of linear size. Additional vertex groups introduce new rays only through crossings between their edges. This yields a quadratic general ray bound and a linear bound in the one-vertex case. The code constructs the diagrams exactly, lifts their overlay vertices directly, and omits the final rank filter because the face theorem proves extremality.

Statement	Scope and status
Canonical lifting and coordinate faces	Recalled from established Q-theory and the preceding project report; reproved here for completeness.
Planar interaction bound	Proved, including degeneracies; sharp for abstract rational affine-potential systems.
Linear one-vertex bound	Proved for matching nullity at most three; an effective-dimension extension is also proved.
Complete enumeration and replay	Implemented with exact rational arithmetic, source binding, limits, and independent reconstruction.
Euler-corner precheck	Proved as an elementary consequence of canonical extension; separately audited and proposed for future runtime integration.
Measured performance	Paired supplied-sector measurements; not a claim of whole-recognizer speedup.
Global recognition from this primitive	Still depends on an unproved coverage and state-control theorem.

The distinction between a supplied sector and all sectors matters. The number of compatible type assignments can grow exponentially with the tetrahedron count even when each individual assignment is easy. Our theorem removes a local enumeration obstacle without identifying an adequate small family of assignments for every unknot.

2 Sources, current state, and the relation to prior work

2.1 The exact repository boundary

The repository was inspected at commit 8188525b70033dcfe7c51ea5ae2c8723ad0c0198, whose timestamp is 2026-10-09 22:03:21 UTC. The maintained implementation includes `fastunknot/nor`

`mal_sector.py`, its separate dense verifier, and compressed normal-component and essential-disc observers. The incoming directory at that commit contains the immediately preceding `unknot_minimum_envelopes_20261009.zip`. Its all-dimensional coordinate-face correspondence and nullity-two implementation are the direct starting point for this continuation.

Other same-day incoming archives concern active-face batched descent, trace search, affine orbit profiles, cycle-envelope completion, and rooted disc components. These are distinct directions. In particular, the phrase “cycle envelope” in another archive refers to a finite boundary-completion problem, not a solution of the planar minimum-subdivision question. The present patch is additive and does not apply an older full-tree snapshot over the newer maintained source.

2.2 Normal coordinates and conversion

Canonical extension from quadrilateral coordinates is classical. Burton defines the canonical extension, describes graph traversal followed by minimum subtraction, and gives a general quadrilateral-to-standard conversion algorithm [1]. His Conjecture 5.16 asks for an output-polynomial bound for that general conversion algorithm. Our result addresses a parameter-restricted sector and is not a proof of that unrestricted conjecture.

The geometric construction also has established ancestry. Normal-fan overlays and Minkowski sums of convex polyhedra are standard objects in exact computational geometry. Fogel, Halperin, and Weibel analyze their extremal complexity; Fogel’s thesis treats exact arrangements, degeneracies, and overlay constructions [4, 5]. We give elementary proofs of the planar estimates required here, so their precise normal-sector specialization can be assessed without transferring a theorem between different ambient settings.

One-vertex triangulations are a particularly relevant setting for the linear bound. The preprocessing in Burton–Ozlen’s branching algorithm can produce a one-vertex triangulation without increasing the number of tetrahedra under its stated hypotheses [2]. Nothing here asserts that this preprocessing forces the selected quadrilateral kernel to have dimension three.

2.3 The 2026 hierarchy paper

Lackenby’s July 2026 preprint gives a new hierarchy algorithm, subsequent refinements, and compressed operations under explicit topological hypotheses [6]. Section 9 initially bounds the number of visits by $L(g + 1)^L$, in terms of hierarchy length and maximum pattern complexity. The refined hierarchy has length at most $4c_q(H)$ (Proposition 10.2). Proposition 14.2 gives polynomial compressed cutting and bundle cleanup for uniform-type handle structures on compact orientable irreducible manifolds with essential boundary pattern, along a connected properly embedded normal surface; its size parameters are $|H|$ and the logarithm of extended surface weight. These results provide structural tools for a global complexity analysis. The 2021 lecture announcement is a distinct source[7]; the present paper proves the supplied-sector bound.

3 Exact sector coordinates and the contracted graph

3.1 A compatible sector

Let $\mathcal{T}$ be a finite triangulation of a compact three-manifold with t tetrahedra. The native implementation validates its encoded face pairings and uses finite standard normal coordinates. There are

four triangle types and three quadrilateral types per tetrahedron. A *compatible sector* Σ allows at most one quadrilateral type in each tetrahedron. Let k be the number of allowed types, and write $q \in \mathbb{R}^k$ for their coordinates; all other quadrilateral entries are zero.

The standard sector cone $\mathcal{S}_\Sigma$ consists of the nonnegative triangle and allowed quadrilateral coordinates satisfying the normal matching equations. Because the quadrilateral constraints have been fixed, this is an ordinary pointed rational polyhedral cone. We enumerate its primitive integer extreme-ray generators with nonzero quadrilateral part. The omitted rays are precisely the vertex-link rays.

Across a paired face corner, matching has the form

$$T_b - T_a = \delta_e(q), \quad (1)$$

where δ_e is an integer linear form involving the allowed quadrilateral coordinates on the two sides. Build the triangle difference graph and contract every edge whose label is identically zero on the chosen coordinate sector. Each retained graph vertex is a class of triangle corners which must have equal coordinates.

Write a for the number of retained nonzero-labelled edges, counting loops and repeated edges. Each allowed quadrilateral has four face arcs, so

$$a \leq 4k. \quad (2)$$

Loops must be retained: after contraction, a loop can impose a nontrivial linear condition on q even though its triangle difference is zero.

For every global triangulation vertex with more than one contracted class, write G_v for its class set. Let g be the number of these groups and $p = \sum_v |G_v|$. Singleton groups contribute independent vertex-link coordinates and possibly cycle equations but no minimum subdivision.

3.2 Potentials, cycle constraints, and the relevant dimension

Integrate the labels in (1) along a spanning forest. This gives integer potential forms $\ell_c(q)$ for the triangle classes. The non-tree edges, including loops, give a homogeneous cycle system

$$A_\Sigma q = 0.$$

Let

$$E = \ker A_\Sigma, \quad d = \dim E, \quad \mathcal{C} = E \cap \mathbb{R}_{\geq 0}^k. \quad (3)$$

Within a vertex group, all triangle solutions have the form $T_c = \alpha_v + \ell_c(q)$. The potentials depend on the spanning forest, but changing them by a common form in a group does not change any minimum region or canonical lift.

Lemma 3.1 (Cycle-rank count, recalled). *The contracted graph satisfies*

$$p - g \leq a - k + d \leq 3k + d.$$

In particular, for $d \leq 3$, $p - g \leq 3k + 3$ and $p \leq 6k + 6$.

Proof. Include singleton groups when counting vertices and connected components of the retained graph. They add equally to those two counts. Its cycle space therefore has dimension $a - p + g$. Every independent row of the quadrilateral cycle system comes from that cycle space. Since $\text{rank } A_\Sigma = k - d$, we have $k - d \leq a - p + g$. Rearranging and using (2) proves the first two inequalities. Every retained group has at least two classes, so $g \leq p/2$ and $p \leq 2(p - g)$. $\square$

The measured implementation stores a rational basis of E and its integer graph potentials. Its standard kernel has at most $8k$ triangle variables and k quadrilateral variables before using the refined dimension-specific count above. The original triangulation remains available for full coordinate expansion, geometric validation, and disc testing.

4 Canonical faces and projective minimum subdivisions

4.1 Canonical lifting

For $q \in \mathcal{C}$, set

$$m_v(q) = \min_{c \in G_v} \ell_c(q), \quad \mathcal{L}(q)_c = \ell_c(q) - m_v(q).$$

The quadrilateral part of $\mathcal{L}(q)$ is q . Singleton groups have canonical triangle coordinate zero. Let L_v be the standard vertex-link vector at the global vertex v .

Lemma 4.1 (Canonical decomposition). *Every $z \in \mathcal{S}_\Sigma$ has a unique expression*

$$z = \mathcal{L}(q) + \sum_v \beta_v L_v, \quad \beta_v \geq 0,$$

where q is its quadrilateral projection and β_v is its minimum triangle value at v . Every non-link extreme ray is canonical.

Proof. Matching fixes the triangle differences. Subtracting the minimum value in each group gives the displayed canonical triangle values; what remains is exactly a nonnegative multiple of the corresponding link. This also proves uniqueness. If $q \neq 0$ and some $\beta_v > 0$, the canonical part and that link are nonzero, nonproportional vectors of the cone, so their sum cannot span an extreme ray. If $q = 0$, the independent nonnegative link factors give exactly the link rays. $\square$

For the face construction, also include each omitted singleton group and use its unique class as its anchor. For anchors $b_v \in G_v$, define

$$F_b = \{q \in \mathcal{C} : \ell_c(q) \geq \ell_{b_v}(q) \text{ for all } v, c \in G_v\}. \quad (4)$$

Theorem 4.2 (Coordinate-face correspondence, recalled). *On F_b , canonical lifting is a linear isomorphism onto the coordinate face of $\mathcal{S}_\Sigma$ on which the anchor triangle at every global vertex is zero. Consequently the non-link standard extreme rays are exactly the lifts of the extreme rays of the cells F_b , with duplicates removed.*

Proof. The selected anchor attains the minimum on F_b , so the canonical triangle entries are the linear forms $\ell_c - \ell_{b_v}$. Conversely, a matching vector with those anchor entries zero has exactly those triangle coordinates; their nonnegativity is equivalent to (4). Quadrilateral projection and this linear lift are inverse.

Setting nonnegative coordinates equal to zero defines a face of the entire standard cone. An extreme ray of that face is extreme in the entire cone. Conversely, choose a zero triangle at each vertex of a canonical non-link extreme ray. It belongs to the corresponding anchor face and remains extreme there. This proves both directions without assuming global additivity of canonical lifting. $\square$

This theorem is already present in the preceding minimum-envelope report [8]. Its proof is included because the subsequent counting and rank-free enumeration depend on the precise face assertion. An equality between two potentials *above* the minimum does not in general define such a face; this is why the full equality arrangement needs a rank filter.

4.2 Normalization and degeneracies

Normalize every nonzero quadrilateral ray by

$$P = \{q \in \mathcal{C} : \mathbf{1}^T q = 1\}. \quad (5)$$

This is a compact rational polytope because it lies in the unit simplex. For $d \leq 3$, it is empty, a point, a segment, or a two-dimensional convex polygon. It can have smaller dimension than $d - 1$.

Corollary 4.3. *The non-link standard rays are exactly the primitive canonical lifts of the vertices of the common minimum subdivision of P .*

Proof. The functional $\mathbf{1}^T q$ is strictly positive on every nonzero vector of an anchor face, since that face has no remaining pure link direction. Its unit section meets each ray once. Vertices of that section correspond to its extreme rays; apply Theorem 4.2. $\square$

For example, a three-dimensional linear kernel can intersect the nonnegative orthant only in a ray. Checking rational nullity alone would not justify using a two-dimensional polygon routine without its point and segment branches. Both delivered implementations classify the actual section exactly. On a segment, the relevant vertices are its endpoints and true breakpoints of the one-dimensional minimum envelopes. A point gives one non-link ray; an empty section gives none.

5 A sharp planar interaction theorem

Assume now that P is two-dimensional, and let b be its number of corners. Restrict each potential to $\text{aff } P$, where it becomes an affine function in two variables. For group v , identify identical restricted functions and let f_v be the number whose minimizing regions have nonempty two-dimensional relative interior in P . Put

$$A_v = f_v - 1, \quad A = \sum_v A_v. \quad (6)$$

Each $f_v \geq 1$. A group with $f_v = 1$ contributes no internal subdivision and has $A_v = 0$.

Lemma 5.1 (Discarding zero-area minimizers). *Removing a restricted affine function whose minimum region has no two-dimensional relative interior does not change the lower envelope or introduce a missing subdivision vertex. Its original triangle coordinates must still be reconstructed when lifting.*

Proof. In the interior of P , avoid the finitely many equality lines between distinct functions. The minimizer is then unique and remains so in an open neighborhood. Such points are dense in P . Passing to a subsequence of minimizing labels and using continuity shows that the retained functions give the same minimum at every point of P , including its boundary.

On a relative-open face of the resulting subdivision, the difference between a discarded function and an attaining affine function is affine and nonnegative. If it vanishes at a relative-interior point

of that face, it vanishes identically on its affine hull: otherwise it would take both signs nearby. Thus a discarded function cannot introduce an isolated extra vertex inside such a face. Removing it from the geometric calculation does not remove its place in the full triangle output. $\square$

5.1 One minimum diagram

Lemma 5.2 (Linear size of one diagram). *A minimum diagram with f full-dimensional cells has at most $2(f - 1)$ vertices other than the original corners of P , and at most $3(f - 1)$ interior edges, after unnecessary collinear subdivisions are merged.*

Proof. Every minimum cell is the intersection of P with affine halfplanes, so it is convex. Let I be the number of interior diagram vertices, B the additional boundary vertices, and e the interior edges. Its total vertex count is $b + B + I$ and its boundary-edge count is $b + B$. Euler's formula for the connected planar subdivision gives

$$e = I + f - 1. \quad (7)$$

After collinear degree-two vertices are suppressed, every interior vertex has at least three incident interior edges. A degree-two bend cannot occur: two neighboring affine cells share a straight equality line. Every new boundary vertex has at least one incident interior edge. Counting interior-edge ends therefore gives

$$2e \geq 3I + B. \quad (8)$$

Edges meeting original polygon corners add nonnegative incidences and only strengthen this inequality. Combining (7) and (8) yields $I + B \leq 2(f - 1)$ and then $e \leq 3(f - 1)$. $\square$

5.2 Two diagrams

Lemma 5.3 (Crossing bound). *For two minimum diagrams with f_1 and f_2 full-dimensional cells, the number X of proper transverse crossings between their open interior edges is at most*

$$X \leq (f_1 - 1)(f_2 - 1).$$

No general-position hypothesis is required.

Proof. First suppose the diagrams are in general position with respect to each other. Every proper crossing adds one interior overlay vertex and splits each of two edges. Applying (7) to the two diagrams and their overlay gives

$$F = f_1 + f_2 - 1 + X,$$

where F is the number of two-dimensional overlay cells. Each overlay cell is the intersection of one cell from each diagram. Convexity makes that intersection connected, so $F \leq f_1 f_2$, proving the inequality.

For the general case, first remove duplicate functions and zero-area minimizers as in Lemma 5.1. Each retained function has a strict minimizing witness in the interior of P . All these finitely many strict inequalities persist under sufficiently small perturbations of the affine coefficients, so the same f_1, f_2 functions still have full-dimensional cells.

At a proper crossing of open edges, each diagram has precisely two adjacent retained affine pieces locally. Three distinct functions which agree along a whole open edge have collinear differences

normal to that line; an intermediate function cannot be strictly least on either side and would not have been retained. Strict gaps from all other retained pieces and the transversality of the two lines persist under small perturbations. Choose disjoint neighborhoods of the finitely many original crossings. A sufficiently small generic rational perturbation preserves a distinct crossing in each neighborhood while eliminating nongeneric coincidences between the diagrams. Applying the general-position bound to the perturbed diagrams bounds the original crossings as well. $\square$

5.3 The complete bound and its normal-surface specialization

Theorem 5.4 (Planar interaction bound). *The complete number R of non-link standard extreme rays satisfies*

$$R \leq b + 2 \sum_v A_v + \sum_{u < v} A_u A_v, \quad (9)$$

$$R \leq b + 2A + \binom{A}{2} = b + \frac{A(A+3)}{2}. \quad (10)$$

Proof. By Corollary 4.3, count the vertices of the common minimum subdivision. Every such vertex is either an original corner of P , a vertex of one individual diagram, or a proper transverse crossing of open edges from distinct groups. A collinear overlap contributes only endpoints already belonging to the individual diagrams. An intersection at an individual diagram vertex is already counted there. If more than two groups cross at a new point, any two nonparallel incident edges witness that point, and counting it more than once only enlarges the bound.

Lemma 5.2 bounds the individual vertices by $2 \sum_v A_v$; Lemma 5.3 bounds the remaining crossings by $\sum_{u < v} A_u A_v$. This proves (9). Since the A_v are nonnegative integers,

$$2 \sum_{u < v} A_u A_v = A^2 - \sum_v A_v^2 \leq A^2 - A,$$

which gives (10). $\square$

Corollary 5.5 (Quadratic normal-sector bound). *For a sector of matching nullity $d = 3$, with a genuine polygon section,*

$$R \leq k + \frac{(3k+3)(3k+6)}{2} = \frac{9k^2 + 29k + 18}{2}.$$

In particular $R = O(k^2)$.

Proof. The polygon P is cut out in its affine plane by the k coordinate inequalities $q_i \geq 0$, so $b \leq k$. Also $f_v \leq |G_v|$ and $A \leq p - g \leq 3k + 3$ by Lemma 3.1. Substitute in Theorem 5.4. $\square$

Corollary 5.6 (Linear one-vertex bound). *If at most one group has $f_v > 1$, the interaction term vanishes and*

$$R \leq b + 2A \leq 7k + 6.$$

This holds in particular for a one-vertex triangulation with matching nullity at most three, with the empty, point, and segment cases included.

Proof. For a polygon, apply (9). For a segment, each distinct active affine function contributes at most one open interval of the lower envelope. There are at most $p - g$ breakpoints in total, giving $R \leq p - g + 2 \leq 3k + 5$ even when the raw nullity is three. Empty and point sections give zero and one rays, respectively. These satisfy the displayed looser bound. $\square$

The constants in the normal-sector corollaries are deliberately conservative. The data-dependent expression (9) is usually much smaller. The stronger one-vertex conclusion is structural: there is no second minimum diagram to create a crossing. It does not follow merely from a small measured output list.

6 Sharp examples and effective dimension

6.1 Exact sharpness in the rational-potential model

Theorem 6.1 (Sharp abstract interaction count). *For every finite prescribed list of nonnegative integers A_v , the bound (9) is attained by rational affine-potential systems on a rational convex polygon.*

Proof. Choose a rational convex polygon with 0 in its interior and pairwise nonparallel nonzero rational normals n_v . For group v , draw A_v parallel lines $n_v \cdot x = \tau_{v,s}$ with strictly increasing rational offsets. Choose all offsets sufficiently close to zero that every cross-group intersection lies in the interior of the polygon. This is possible because there are finitely many direction pairs. Choose the offsets generically so that no three lines meet, no line passes through a polygon corner, and no two boundary endpoints coincide.

These strip partitions are minimum diagrams. With $z = n_v \cdot x$, take

$$h_{v,j}(x) = -jz + \sum_{s=1}^j \tau_{v,s}, \quad 0 \leq j \leq A_v.$$

The difference between consecutive functions is $-z + \tau_{v,j}$. On the strip between successive offsets, summing these differences shows that the corresponding $h_{v,j}$ is least. Thus all $A_v + 1$ pieces are active in succession.

The common subdivision has exactly b original corners, $2 \sum_v A_v$ distinct boundary endpoints, and $\sum_{u < v} A_u A_v$ distinct interior crossings. These exhaust its vertices and establish equality. $\square$

Example 6.2 (Two independent strip families). On $[0, 1]^2$, take

$$h_j(x, y) = -jx + \frac{j(j+1)}{2(m+1)}, \quad 0 \leq j \leq m,$$

and the analogous $n + 1$ functions in y . Their break lines are $x = j/(m + 1)$, $1 \leq j \leq m$, and $y = \ell/(n + 1)$, $1 \leq \ell \leq n$. The overlay has exactly $(m + 2)(n + 2) = mn + 2m + 2n + 4$ vertices.

Rational affine functions on a normalized section can be homogenized using $\mathbf{1}^T q$ and then cleared of denominators. Thus these examples are genuine rational potential cones. They do not automatically satisfy the local incidence restrictions of a triangulated three-manifold. We have not proved that the quadratic lower bound is realizable by such triangulations, or that the constants in Corollary 5.5 are sharp there.

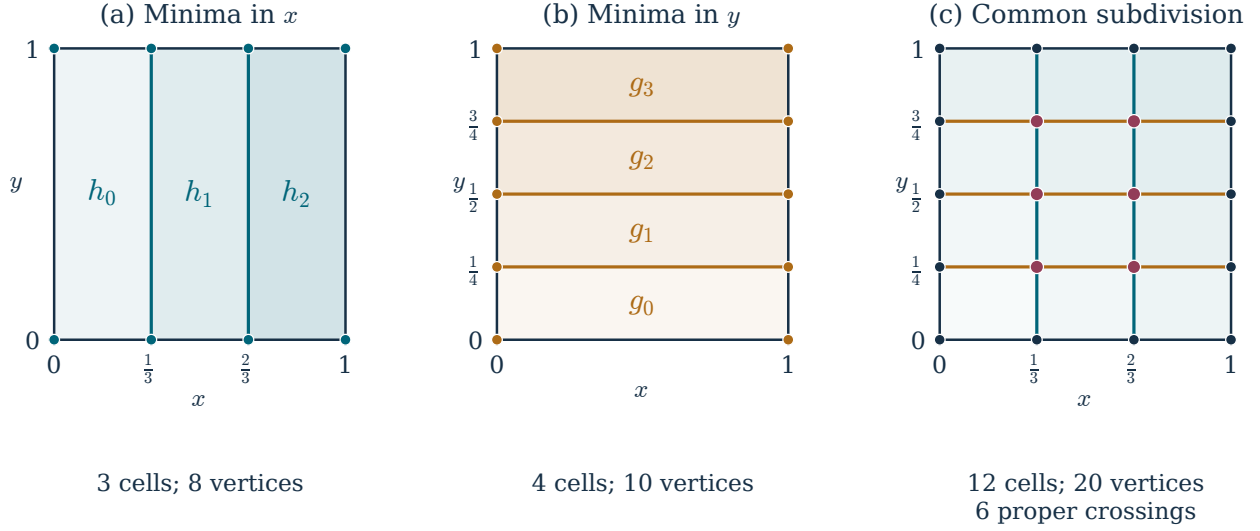


Figure 1: Two minimum diagrams and their common subdivision. Each individual diagram has linear size. The proper intersections between groups generate the quadratic term. The illustration is an exact rational-potential example, not a claimed normal-sector realization of a knot exterior.

6.2 An extension by removing forced-zero coordinates

The implemented admission condition is $d \leq 3$. A stronger mathematical condition uses the dimension of the feasible cone itself.

Proposition 6.3 (Effective-dimension reduction). *Let J consist of those quadrilateral coordinates which are positive in at least one vector of $\mathcal{C}$. Remove the other allowed types and rebuild the sector. Its matching kernel equals the linear span of its nonnegative cone. Consequently, if $\dim \text{span } \mathcal{C} \leq 3$, the rebuilt sector satisfies the nullity-at-most-three condition and has the same non-link standard rays.*

Proof. Deleting coordinates which are identically zero on $\mathcal{C}$ changes no feasible normal vector. By exactness of the rebuilt cycle system and canonical lifting, its nonnegative kernel is exactly the quadrilateral projection of that unchanged feasible standard cone. For each $j \in J$, choose a nonnegative feasible vector positive at coordinate j and sum these finitely many vectors. The resulting vector is strictly positive in every surviving coordinate. Within the surviving matching kernel, a sufficiently small perturbation of this vector in any direction remains nonnegative. Hence the feasible cone contains a relative-open neighborhood in that kernel and spans it. The dimensions agree. Recompile preserves the same feasible standard cone, so the claimed ray correspondence follows. $\square$

The active support can be identified with a single linear program, using the following standard maximal-support construction specialized to the homogeneous matching cone. Its dual also gives a compact independently checkable certificate for the proposed preprocessing.

Corollary 6.4 (One-LP support extraction and its certificate). *For $\mathcal{C} = \{q \geq 0 : A_\Sigma q = 0\}$, consider*

$$\max \mathbf{1}^T z \quad \text{subject to} \quad A_\Sigma q = 0, \quad q \geq 0, \quad 0 \leq z \leq \mathbf{1}, \quad z \leq q. \quad (11)$$

Its optimum is $|J|$, and every optimal z is the indicator vector $\mathbf{1}_J$. The associated q is positive on every coordinate in J . Moreover J is exactly the possible positive support if there exist rational vectors q, y such that

$$\begin{aligned} A_\Sigma q &= 0, & q &\geq 0, & q_j &\geq 1 \quad (j \in J), \\ w &= A_\Sigma^T y, & w_j &= 0 \quad (j \in J), & w_j &\geq 1 \quad (j \notin J). \end{aligned}$$

Such witnesses always exist for the true J and can be obtained from exact primal and dual optima. Their verification requires only rational matrix arithmetic and inequalities.

Proof. Outside J , every feasible q_j , and hence z_j , is zero. Thus the objective is at most $|J|$. For each $j \in J$, scale a feasible vector positive at j until that coordinate is at least one. The sum of those vectors has every coordinate of J at least one. Taking $z = \mathbf{1}_J$ attains $|J|$. Equality in the upper bound forces every optimal z to be exactly this indicator.

The dual of (11) is

$$\min \mathbf{1}^T u \quad \text{subject to} \quad u, v \geq 0, \quad u + v \geq \mathbf{1}, \quad A_\Sigma^T y \geq v, \quad y \text{ unrestricted.}$$

For any feasible dual, $w = A_\Sigma^T y \geq 0$. Since $w^T x = 0$ for every $x \in \mathcal{C}$, a positive feasibility witness at $j \in J$ forces $w_j = v_j = 0$ and therefore $u_j \geq 1$. Strong duality gives an optimum with $\mathbf{1}^T u = |J|$, which forces $u_j = 1$ on J and $u_j = 0$ elsewhere. The inequalities then give $w_j \geq v_j \geq 1$ off J . The primal construction and this dual vector give the displayed witnesses.

Conversely, supplied witnesses q, y prove both inclusions without trusting a solver status. The first row proves that every $j \in J$ can be positive. For any feasible x , the second row gives $0 = y^T A_\Sigma x = w^T x$ with nonnegative terms and strictly positive weights off J . Therefore every x_j off J is zero. These are exactly the two required support assertions. $\square$

The program in (11) uses the unnormalized homogeneous cone. Adding $\mathbf{1}^T q = 1$ or an arbitrary coordinate cap would invalidate the scaling argument. Exact rational linear programming supplies this preprocessing in polynomial bit time. This is a mathematical extension, *not implemented* in the present branch. The 188 selected corpus sectors of higher raw nullity are reported as outside the implemented gate, even if some could benefit from this reduction.

7 The exact producer and its complexity

7.1 Constructing a bounded projective chart

Let $b_0, \dots, b_{d-1}$ be a rational basis of E , and set $s_i = \mathbf{1}^T b_i$. If every $s_i = 0$, the nonnegative cone has no nonzero vector. Otherwise choose $s_h \neq 0$ and define

$$u = b_h / s_h, \quad w_i = b_i - s_i u \quad (i \neq h).$$

Then every point of the affine matching section is uniquely $q = u + \sum_{i \neq h} x_i w_i$. For $d = 3$, this is a two-variable rational chart; each coordinate inequality and each potential is affine in the chart variables.

No guessed bounding box is used. Two quadrilateral-coordinate rows are independent on the two-dimensional direction space. Inverting those rows maps their possible values in $[0, 1]^2$ to a rational parallelogram in the chosen chart. The implementation takes its containing axis-aligned rectangle and clips it by all inequalities $q_j \geq 0$. Every feasible point lies in that rectangle, and the result is exactly P . An equivalent implementation could use those two quadrilateral coordinates as its chart and start directly from $[0, 1]^2$.

Closed halfplane clipping preserves convex cyclic order. Consecutive duplicates and redundant collinear vertices are removed. A collinear result becomes its two endpoints, and a singleton is preserved. Thus the same construction detects every feasible dimension without numerical tolerances.

7.2 Minimum cells and overlay points

For each group and each distinct projected form h_i , clip P by $h_j - h_i \geq 0$ for all other forms in that group. Retain the resulting polygons with positive area. Their shared interior edges are genuine minimum-diagram edges. Edges on the boundary of P are not copied into the interior overlay stage; the polygon's own corners are included once.

The output point set consists of the polygon corners, all interior-edge endpoints, and all nonparallel intersections of segments belonging to different groups. An intersection is accepted only when it lies in both closed segments. Coincident segments are merged; a collinear overlap requires no extra point beyond its endpoints. Group ownership allows the implementation to skip segment pairs from a common subdivision.

The point and segment branches use their actual feasible affine hull. For a segment, an exact sorted-slope lower-envelope stack finds all genuine minimum changes, including coincident lines and endpoint-only ties. No positive-dimensional equality above a minimum is introduced.

Theorem 7.1 (Producer correctness). *With its allowances disabled, the delivered planar producer terminates for every valid supplied sector of matching nullity at most three and returns exactly its primitive non-link standard rays. It requires neither a final supported-rank test nor enumeration of all potential equality intersections.*

Proof. The chart construction gives exactly P . Each clipped region is exactly the region where its nominated potential is minimal. Lemma 5.1 justifies retaining only full-dimensional regions in the polygon branch. The vertex classification in the proof of Theorem 5.4 shows that the point set is the complete common-subdivision vertex set, with no spurious subdivision directions. The exact segment envelope proves the same claim in dimension one. Corollary 4.3 then establishes standard extremality and completeness. Primitive integer normalization and the canonical lift give the required generators. $\square$

7.3 Preprojected lifting

It would be wasteful to reevaluate every k -coefficient potential on every output vector. The implementation projects each potential once to its constant term and at most two chart coefficients. At an output point, compute the rational normalized q , clear denominators, and divide by its coordinate gcd. If the resulting primitive vector is $\hat{q}$, its scale from the normalized section is $\mathbf{1}^T \hat{q}$. Evaluate each projected potential, subtract the minimum within its group, and multiply by that scale. Finally copy each contracted triangle value to its original corners.

The graph potentials are integer forms in q . Hence an integer $\hat{q}$ gives an integral canonical lift. Moreover a primitive quadrilateral vector gives a primitive full standard vector. Conversely, if all quadrilateral entries of a canonical standard vector had a common factor, all its triangle entries would have that factor as well. Primitive normalization in quadrilateral coordinates is therefore correct here.

Theorem 7.2 (Conservative arithmetic and bit bounds). *After the supplied sector kernel has been built, complete enumeration can be performed with*

$$O(1 + k^3 + (t + k)R)$$

rational arithmetic operations, with polynomial storage and exact deterministic set bookkeeping. The full bit complexity, including rational kernel construction and coordinate output, is polynomial in the encoded input size for $d \leq 3$. The cubic expression is an arithmetic bound, not a cubic bit-time claim.

Proof. By Lemma 3.1, $p = O(k)$. Preprojecting all potentials takes $O(pk)$ arithmetic operations. Constructing the section by successive clipping takes $O(k^2)$: after j halfplanes its polygon has at most $j + 4$ corners. For a group of p_v distinct forms, clipping all its candidate cells costs at most $O(p_v^2(k + p_v))$, since each intermediate polygon has at most $k + p_v + 4$ sides. Summing gives $O(k^3)$.

The genuine interior skeleton has at most $3A = O(k)$ segments. Pairwise intersection therefore costs $O(k^2)$. Boundary recognition and duplicate management remain within the cubic bound. One may deterministically collect, sort, and merge rational records instead of assuming constant-time hashing; the additional $O(k^2 \log(k + 2))$ comparisons are absorbed by the cubic estimate. The cubic bound applies to geometric arithmetic together with this deterministic sort-and-merge realization. The provided Python implementation uses exact dictionaries and sets. Its worst-case hash-collision cost is not included in that sharper bound; its worst-case bit complexity remains polynomial. For example, collecting $O(k^2)$ points in a hash set could require $O(k^4)$ equality comparisons in an adversarial collision pattern.

With preprojection, each full lift costs $O(t + p + k) = O(t + k)$ operations. The inherited native coordinate check examines a linear-size family of local matching and coordinate records. This yields the displayed total.

For bit complexity, graph paths give integer coefficients of polynomial bit length. Rational elimination has polynomial coefficient growth by determinant bounds. Each reduced chart coefficient has polynomial bit length; a polygon vertex is an intersection of two affine boundary lines. Its reduced numerator and denominator are bounded by fixed-size determinants of those coefficients. This remains true under repeated clipping, since a retained vertex is always supported by original boundary lines. Segment intersections satisfy the same fixed-dimensional determinant bound. Integer normalization, sorting, and full output use polynomial-size integers. Thus exact bit complexity is polynomial; kernel construction and the arithmetic cost of large integers are not hidden inside a claimed unit-cost full running time. $\square$

Lemma 7.3 (Inherited height bound). *If a primitive non-link standard ray actually uses $s \geq 1$ quadrilateral types, every triangle coordinate is at most 4^s and each quadrilateral coordinate at most 4^{s-1} . In particular each output coordinate has $O(k)$ bits.*

Proof. In the retained matching matrix $[B \mid D]$, the triangle block B is a graph incidence block and is totally unimodular. Each quadrilateral column of D has absolute column sum at most

four. Expanding a mixed minor successively in its quadrilateral columns bounds its absolute value by 4 to the number of such columns. On an extreme positive support, the restricted kernel is one-dimensional. Select an independent spanning set of its matrix rows. Its primitive generator is then given by maximal cofactors divided by their gcd. Deleting a triangle column leaves at most s quadrilateral columns; deleting a positive quadrilateral column leaves at most $s - 1$. Repeating contracted triangle values in the original triangulation changes neither bound. $\square$

This coordinate bound controls encoding, not the number of elementary normal pieces. The latter can be exponentially larger. The implementation never expands those pieces.

8 Independent completeness checking

8.1 A different reconstruction

The independent checker imports neither the planar producer nor the contracted sector producer. It starts from the original dense standard matching equations, eliminates the triangle columns independently, and constructs another rational basis and another projective chart. Its chart deliberately uses a different pivot convention. Complete primitive quadrilateral directions, which are chart-independent, are compared.

For each group and each pair of distinct restricted affine forms h_i, h_j , the checker imposes

$$h_i = h_j, \quad q_s \geq 0 \ (1 \leq s \leq k), \quad h_r - h_i \geq 0 \text{ for all } r \text{ in that group.} \quad (12)$$

The nonzero equality is a line. Parameterize it rationally and restrict all inequalities to that line. Every inequality supplies a lower bound, upper bound, constant constraint, or contradiction. The result is empty, a point, or a closed bounded segment. This is interval intersection, not polygon clipping or the producer's envelope traversal.

The checker retains the endpoints of all feasible ties, omits boundary segments from the interior crossing stage, merges identical segments, and intersects the remaining segments across distinct groups. It adds the independently computed polygon corners. Point and segment sections are verified in their actual dimension.

Lemma 8.1 (Minimum-tie reconstruction). *A positive-length feasible tie in (12) away from ∂P is a genuine maximal minimum-diagram edge. A singleton feasible tie is a diagram vertex. After duplicate removal, the method recovers exactly the individual minimum skeletons and their vertices.*

Proof. At a feasible point both forms attain the group minimum. Along a feasible tie segment, every third affine gap is nonnegative. If that gap vanishes at a relative-interior point of the segment, its restriction to the line must vanish identically. Thus the active minimum structure cannot change at an isolated interior point of the segment. A positive tie interval is therefore not an artificial proper subdivision of a ridge.

If a feasible tie is a singleton, its equality line meets another active constraint in an independent direction; otherwise a nontrivial interval would remain. That constraint is either a coordinate boundary or another minimum equality. Hence the point is a diagram vertex. Boundary tie segments add only their endpoints, which are existing diagram vertices or polygon corners. Conversely, every genuine edge separates two minimum pieces and is obtained from their equality; all its endpoints are retained. The same reasoning covers redundant forms that win only along an edge or at a point. They can reproduce an existing feature but cannot invent one. $\square$

Theorem 8.2 (Independent ray-list certification). *For a valid supplied triangulation and a sector with $d \leq 3$, successful completion of the independent checker certifies that its source-bound primitive quadrilateral list is exactly the complete non-link standard ray list. A missing ray, additional ray, repeated ray, nonprimitive ray, or source mismatch is rejected. A resource interruption supplies no successful verification.*

Proof. Independent dense elimination describes the same matching sector. Lemma 8.1 and the vertex classification in Theorem 5.4 reconstruct its full projective vertex set. Corollary 4.3 identifies the exact standard rays. Normalization gives a unique primitive quadrilateral vector per ray. The checker validates the schema, exact source hash, compatible type list, integer signs, gcds, ordering, and equality of the complete sets. Limits raise an interruption rather than returning success. $\square$

The independent dense model first projects its $4t$ triangle-potential rows and deduplicates the results, costing $O(tk)$ arithmetic operations. After omitting groups with only one distinct projected form, let p count the remaining distinct forms. Tie construction costs $O(\sum_v p_v^2(k + p_v))$, and the number of distinct internal segments is $O(p)$. The checker uses a cubic coordinate-pair method to construct the section. After the $O(tk)$ projection and deduplication stage, its geometric work is $O(1 + k^3)$ using the contracted count on distinct forms; this statement excludes final length- k output normalization and ordering, and uses the same deterministic-bookkeeping convention as Theorem 7.2. The Python set implementation has the previously stated polynomial collision-cost qualification. Dense elimination is charged separately. Its independent dense model can be more expensive than the producer’s contracted graph. Replay time is measured separately; it is not included in a producer-only timing claim.

8.2 The certificate and its trust boundary

The enumeration schema is `normal-sector-planar-rays-v1`. It contains the hash of the exact supplied triangulation encoding, a sorted compatible type list, and a sorted list of primitive quadrilateral ray vectors. It does not ask a checker to trust supplied areas, floating-point samples, unverified winning labels, or a producer’s traversal log. Full coordinates can be reconstructed independently from each accepted quadrilateral vector.

The source hash is a binding device, not a theorem that the source is a particular knot exterior. Native triangulation validation and the older dense matching model are explicit shared dependencies. The regression suite also disables the producer and tests the checker on already-created certificates, confirming this intended software separation.

9 Disc queries and integration contracts

9.1 What a supplied-sector disc query proves

The new search iterates through complete standard rays and uses the native compressed essential-disc observer on the positive-Euler candidates. The observer’s positive certificates undergo its existing independent replay. This uses the repository’s established compressed normal-component machinery, whose background is Agol–Hass–Thurston orbit processing [3].

Lemma 9.1 (Why the Euler filter is safe on these outputs). *A primitive integral extreme-ray normal surface in a compatible sector is connected. Consequently, a ray with nonpositive Euler characteristic cannot contain a disc component.*

Proof. If the surface were disconnected, the coordinate vectors of its nonempty components would sum to its coordinate vector. Each component is nonnegative, obeys matching, and stays inside the same compatible sector. Extremality makes every component vector proportional to the whole ray. If x is a primitive integer vector and λx is integral, Bezout's identity on the coordinates of x implies $\lambda \in \mathbb{Z}$. A proper nonempty component would require $0 < \lambda < 1$, a contradiction. A connected surface of nonpositive Euler characteristic is not a disc. $\square$

Positive Euler characteristic alone does not certify an essential disc: spheres, projective planes, and inessential discs are possible. Each positive candidate still receives the compressed topological and boundary-essentiality test. A returned witness uses the already-supported `normal-sector-witness-v1` schema. Complete negative search uses `normal-sector-planar-exhaustion-v1`, which combines the full enumeration proof with the necessary negative component certificates.

An exhausted sector excludes the corresponding vertex-disc witnesses in that sector. It is not a conclusion about every sector of the manifold. Nor does a supplied triangulation, by itself, identify an exterior of the diagram that entered the whole recognizer.

9.2 An Euler precheck using only quadrilateral rays

Complete standard-ray enumeration is more work than a query asking only whether a non-link ray has positive Euler characteristic. The following elementary consequence of canonical extension separates these two tasks. Its ingredients are classical: canonical surfaces have no vertex-link components, and canonical quadrilateral vertex surfaces are standard vertex surfaces [1, Section 3 and Lemma 4.5]. The explicit convexity formulation below is used as a proposed optimization; no claim of literature priority is made.

Theorem 9.2 (Objectives nonnegative on links attain a maximum at Q-corners). *Let λ be a linear functional on standard matching vectors such that $\lambda(L_v) \geq 0$ at every global vertex. On the supplied sector cone, $F_\lambda(q) = \lambda(\mathcal{L}(q))$ is positively homogeneous and convex. If the normalized section P is nonempty, then*

$$\max_{q \in P} F_\lambda(q) = \max_{a \in \text{Vert}(P)} F_\lambda(a).$$

Every corner lift $\mathcal{L}(a)$ spans a non-link standard extreme ray. For a compact finite three-manifold triangulation, this applies to Euler characteristic, because each vertex link is a disc or sphere. Consequently there is a non-link standard ray of positive Euler characteristic if and only if some canonical lift of a quadrilateral-cone extreme ray has positive Euler characteristic. The convexity and equivalence hold in all matching dimensions.

Proof. Let $U(q)$ be the signed linear matching vector with triangle entries $\ell_c(q)$ and quadrilateral entries q , with zero offsets in every vertex group. Include the singleton groups with potential zero. Then

$$\mathcal{L}(q) = U(q) - \sum_v m_v(q) L_v, \quad F_\lambda(q) = \lambda(U(q)) - \sum_v \lambda(L_v) \min_{c \in G_v} \ell_c(q).$$

Each minimum of linear functions is concave and positively homogeneous. The coefficients $\lambda(L_v)$ are nonnegative, so the displayed expression is convex and positively homogeneous. If $q = \sum_i \alpha_i a_i$

is a convex combination of corners of P , then

$$F_\lambda(q) \leq \sum_i \alpha_i F_\lambda(a_i) \leq \max_i F_\lambda(a_i).$$

The reverse maximum inequality follows since every corner lies in P .

A corner of P spans an extreme ray of $\mathcal{C}$. Choose any anchor cell containing it. That direction remains extreme in the anchor cell, so Theorem 4.2 makes its lift standard-extreme. Conversely, every non-link standard extreme ray is canonical. Euler characteristic is a homogeneous linear functional in standard normal coordinates [2, Section 2 and Lemma 10], and its values on the links are one or two. Applying the maximum equality and positive rescaling proves the claimed positive-Euler equivalence. $\square$

The comparison concerns the normalized quantity $\chi(\mathcal{L}(q))/(\mathbf{1}^T q)$ when primitive integer directions are used. Primitive normalization can change the value of Euler characteristic by a positive scale; it does not change its sign. Unnormalized numerical maxima of primitive representatives must not be substituted for the maximum in the theorem.

Corollary 9.3 (A negative Euler answer excludes essential normal discs). *If all canonical corner lifts have nonpositive Euler characteristic, there is no essential normal disc anywhere in the supplied sector.*

Proof. An essential normal disc is connected and is not a vertex-link disc: the latter has boundary bounding the small disc about a boundary vertex. Hence it has no vertex-link component and is canonical [1, Definitions 3.1 and 3.9, Lemma 3.5]. Its nonzero quadrilateral vector would give $F_\chi > 0$, contradicting Theorem 9.2. The empty-section case has only vertex-link surfaces and gives the same conclusion. $\square$

A positive answer has a different meaning. The returned standard ray may be a sphere, a projective plane, or an inessential disc. Its essentiality still requires the compressed topological observer. The theorem does not assert that an essential disc must itself occur among the canonical quadrilateral corner lifts. It also does not claim that all standard rays occur there; the additional overlay vertices remain necessary for complete enumeration.

Theorem 9.4 (Quadratic post-kernel Euler precheck). *For matching nullity at most three, the positive-Euler non-link ray query can be decided after kernel construction with*

$$O(1 + t + k^2)$$

rational arithmetic operations using deterministic array bookkeeping. A positive answer can include one primitive full-coordinate standard ray within the same bound. This is an arithmetic bound with polynomial bit complexity; compressed component testing and independent dense replay are separate costs.

Proof. First build a linear Euler functional $\chi(z) = e^T z$ in $O(t)$ operations. Count one face for each elementary normal disc, count normal arcs on one representative of each physical triangulation face, and count intersection points on one representative local edge for each global triangulation edge. Matching makes the representative choices consistent. The formula $F - E + V$ yields integer coefficients between -3 and 5 : a normal disc meets at most four faces and four edges. The

face and edge classes are obtained by traversing linear-size incidence graphs. This is the usual bounded-coefficient Euler construction [2, Lemma 10].

Construct the normalized section by the bounded-chart clipping procedure, without constructing any minimum diagram. This costs $O(k^2)$ and gives at most k corners; the empty, point, and segment cases satisfy the same bound on their number of corners. Preproject the $p = O(k)$ retained potentials to the at-most-two-dimensional chart in $O(k^2)$ operations. One pass through the original triangle corners and allowed quadrilateral entries then aggregates the affine coefficients of $\chi(U(q))$ and the link coefficients $\chi(L_v)$ in $O(t + k)$ operations. These are constant-size chart records indexed by the already-known class identifiers.

At a section corner, evaluate the affine term and each group’s minimum using $O(p + k) = O(k)$ operations. Evaluating all corners therefore costs $O(k^2)$. Theorem 9.2 proves that their maximum decides the query. If a positive corner is found, its primitive canonical full lift costs an additional $O(t + k)$ operations. Exact rational chart and output sizes have the polynomial bit bounds established earlier. $\square$

This Euler precheck is proved and examined by the separate exact-data audit, but is *not installed in the delivered runtime dispatcher or certificate APIs*. A production integration should give its negative answer a certificate checking the complete quadrilateral-corner set and all Euler values, without pretending that this is a complete standard-ray list. The measured producer and benchmark source remain unchanged.

9.3 Public interfaces

Interface	Meaning
<code>enumerate_planar_sector</code>	Build the kernel and return the complete non-link full-coordinate ray list and counters.
<code>sector_planar_rays</code>	Reuse a supplied kernel; yield its complete non-link standard rays.
<code>sector_planar_plan</code>	Expose exact chart, minimum cells, skeleton, and complete projective vertex set.
<code>certify_planar_sector</code>	Produce a complete ray list with the new source-bound enumeration certificate.
<code>verify_planar_sector_certificate</code>	Reconstruct the full list by the independent dense/tie-interval route.
<code>discover_planar_in_sector</code>	Search standard rays using the native essential-disc observer and return a witness or restricted exhaustion.
<code>verify_planar_exhaustion</code>	Check enumeration and every necessary negative disc certificate independently.

The patch adds these modules without changing the recognizer’s default policy or the existing arrangement and support algorithms. This provides a narrow integration point and avoids conflicts with other incoming branches. A caller which has a low-nullity sector can opt into the new API immediately. A future dispatcher may compare structural work estimates, but this report does not install an unmeasured global heuristic.

9.4 Allowances, cancellation, and incomplete work

The whole-query producer and certificate wrappers share a `max_work` allowance across kernel construction, projected geometry, and ray lifting. The lower-level supplied-kernel iterator charges only its own work. Counts are disclosed implementation checkpoints, not clock time or bit operations. A separate `max_orbit_cycles` controls the native component stage. The independent verifier’s allowance applies to its reconstruction geometry; dense-model preparation remains separately accounted and cooperatively cancellable.

Exceeding a producer allowance raises `SearchLimit`; certificate wrappers translate that into `INCONCLUSIVE` and omit a completeness certificate. Arbitrary caller cancellation propagates. False-valued callable objects remain callable and are not replaced by a default. No negative decision is inferred from a partial ray list.

10 Validation on exact geometry and normal surfaces

10.1 Declared corpus and independent reference

The frozen corpus contains 48 supplied triangulations: layered solid tori, boundary-capped and interior-subdivided examples, Pachner variants, finite trefoil and figure-eight exteriors, and connected-sum controls. Each source includes a complete standard-ray list produced with Regina. The selected sectors repeat the predecessor’s declared procedure: all occupied supports, every support of size at most two, every compatible support for at most three tetrahedra, and 64 reproducibly seeded full sectors for each larger source.

A compatible coordinate sector is a coordinate face of the nonnegative standard matching cone. Its extreme rays are therefore exactly the full standard rays whose support lies in that sector. Filtering the complete Regina list is consequently a complete oracle for each selected sector; it is not a comparison only against whichever candidates our code happens to find.

Audit quantity	Completed count
Input triangulations	48
Selected compatible sectors	9,995
Eligible sectors with matching nullity at most three	9,807
Additional nullity-three sectors beyond predecessor	463
Actual polygon sections among those 463	420
Actual segment / point / empty sections among those 463	39 / 3 / 1
Primitive ray occurrences compared across sectors	7,674
Rays in the nullity-three sector cases	2,821
Independent dense/tie-interval replays	607
Altered enumeration certificates rejected	328
Fresh full Regina enumerations of corpus sources	48
Fresh full Regina enumerations of double-cap controls	36
Double-cap disc searches and certificate replays	36

Every eligible sector matched the complete oracle. The 463 new cases include 194 one-vertex and 269 multiple-vertex sources. The 188 remaining selected sectors have raw nullity four or five and were

explicitly outside the implemented gate. We do not count them as solved by the effective-dimension proposition, since that preprocessing is not implemented.

Independent replay covers every nullity-three case and three lower-rank cases per source. Mutations delete a ray, repeat a ray, multiply a primitive ray, or change the source binding. Fresh Regina runs reproduce every frozen source’s complete list and all 36 selected double-cap family lists directly from the recorded face pairings.

10.2 Degenerate and arithmetic controls

The focused producer tests include coincident restricted forms, a form which wins only on a line or at a point, overlapping segments across groups, empty and collapsed three-dimensional cones, exact huge rational basis changes, and callback and allowance boundaries. They also disable `kernel.lift` and `kernel.is_standard_ray` to verify that the new branch uses preprojected lifting and does not silently fall back to the old rank filter.

An independent abstract audit uses a separately written complete equality arrangement and active-constraint-rank oracle. The resulting geometry comparison is separate from the genuine triangulation corpus and tests degeneracies that small geometric fixtures may rarely realize.

The separate abstract audit passed 304 exact potential models with a fixed recorded seed: 68 polygon, 58 segment, 84 point, and 94 empty sections. It compared 609 ray directions against 1,409 feasible arrangement candidates using an independent active-rank filter. The largest output in that audit had 16 rays. These are abstract potential systems, so they are not included in the count of genuine triangulation sectors.

A separate certificate audit passed 96 assertions. It checked the full canonical lift against the independent dense model, source and schema binding, omitted and additional rays, scalar multiples, booleans and floating-point impostors, malformed support, Euler-characteristic fields, disc certificates, and invalid negative statuses. It also checked allowance boundaries: on a recorded query needing 136 charged units, allowances of 1, 68, and 135 returned inconclusive, while 136 completed. The packaged portable replays repeat these checks against the final delivered source; the earlier audit records retain their original source hashes.

The complete maintained native regression suite, with the 22 new producer and certificate tests included, passed **1,344 tests in 155.470 seconds**. The suite’s inherited component diagnostics report 13,846 expanded-sheet topology comparisons and 196,712 marked-cover comparisons. These are existing regression checks which remained green; they are not attributed to the new planar implementation. The complete output is preserved in `results/full_tests.log`. The focused producer tests additionally compare 259 small native sectors against the old complete arrangement route, 20 against the dense route, and 100 seeded abstract models against a separately implemented arrangement oracle. The packaged quick workflow reruns the focused tests, the two independent audits, and the Euler-corner comparison; the pinned full checkout is required to rerun every legacy report-dependent test.

An early certificate wrapper did not charge kernel preparation to its whole-query allowance. The budget audit exposed this discrepancy; the wrapper was corrected before the final regression and packaged replays. A final full corpus run against the delivered snapshot again matched all 9,807 eligible sectors and completed all 48 plus 36 fresh Regina checks. Its record is `results/reproduction/final_corpus_audit.json`.

The separate Euler-corner audit covers the same 607 independently replayed sector cases, including

all 463 of raw nullity three. It compares exact normalized Euler maxima at the original projective corners with those of the complete frozen standard-ray lists, and checks that every corner lift belongs to the complete standard list. Every comparison agrees. The cases comprise 420 polygons, 43 segments, 68 points, and 76 empty sections. Among the nonempty cases, 243 have nonpositive maximum and 288 have positive maximum. The corner rule evaluates 1,419 directions in place of 2,898 complete-list ray occurrences, a strict reduction in 446 cases. The 1,479 avoided evaluations are a finite operation-count comparison, not a measured runtime improvement. This audit deliberately obtains its corners from the existing complete planar plan and uses the baseline canonical lift; it tests the mathematical reduction, not an implementation of the proposed $O(1 + t + k^2)$ precheck. The script and exact records are `experiments/audit_euler_corners.py` and `results/euler_corner_audit.json`.

These experiments test the implementations and their contracts. The proofs, rather than the finite corpus, establish the exact ray bounds and completeness for all inputs satisfying the hypotheses.

11 Paired performance measurements

The benchmark compares complete lists on identical inputs. The Euler-only precheck of Section 9.2 is used in neither timed arm. It separates prebuilt-kernel enumeration from a complete build-and-enumerate call. All completed paired cases require identical canonical ray digests. The implementation’s baseline is the maintained arrangement route at the pinned commit, including its rank test. We report its actual hyperplane and attempted-basis counters rather than estimating them from the worst-case formula.

The principal growing family is a Fibonacci layered solid torus with two boundary tetrahedra attached, with one allowed type in every tetrahedron. Its supplied sector has matching nullity three. The bounded instances undergo fresh Regina ray enumeration; its larger binary-coordinate instances test exact scaling without normal-piece expansion. These are supplied sectors on actual triangulations, not a claim that a difficult knot diagram has been recognized faster end to end.

Small controls include all nine choices of the two cap types, selected sectors in trefoil and figure-eight sources and their subdivisions, connected-sum sources, and empty or one-ray sectors. The benchmark also includes repeated new/new timings as a control for drift and small-call noise. Timings depend on Python, machine load, and coefficient size; the raw replicates and environment record are supplied.

11.1 Protocol, pairing, and completeness

The retained run uses Python 3.12.14 on the recorded Linux environment and contains 23 distinct supplied-sector cases, 46 case–protocol combinations, and 216 complete measured arrangement/planar pairs. Each small combination has five measured rounds; the principal family at $n \geq 16$ has three. A round randomly orders the arrangement arm, the planar arm, and a second identical planar arm. Each protocol also records an excluded warmup for each primary arm. Source construction, expected-list derivation, ray hashing, independent coverage replay, and result serialization are outside the timed intervals. Native coordinate validation remains inside enumeration.

There are 752 retained arm calls including warmups: 750 complete and 2 resource-capped. All complete outputs agree with their case’s canonical primitive-ray digest. Separately, the independent exhaustive checker returns `VERIFIED` for all 23 benchmark cases, including the $n = 32$ case whose

arrangement warmups are capped. This replay is outside both enumeration protocols. The additional high-size capability runs below have their own, separately reported replay outcomes.

For complete matched rounds i , the reported ratio is

$$\rho = \text{median}_i \left(\frac{T_i^{\text{arrangement}}}{T_i^{\text{planar}}} \right).$$

The time columns show the separate median durations of the two arms. Their quotient need not equal ρ : pairing is performed before taking the median. Values above one favor the new enumerator. All individual ratios and durations are retained; no ratio is assigned when the baseline does not finish.

11.2 The growing family and the removed work

n	t	Prepared kernel			Build + enumerate			Pairs
		Old (ms)	New (ms)	ρ	Old (ms)	New (ms)	ρ	
1	3	3.026	1.718	1.82	2.894	1.735	1.71	5
2	4	7.921	2.045	3.84	7.818	2.358	3.59	5
4	6	44.521	2.931	15.18	46.822	3.763	12.10	5
8	10	490.356	5.076	95.15	491.897	6.269	78.16	5
16	18	10,219.114	11.955	924.16	9,648.874	15.511	626.20	3
32	34	capped	30.358	—	capped	50.340	—	0

Table 1: Principal double-cap family, with cap types $(1, 1)$ and $t = n + 2$ tetrahedra. Every complete list has seven non-link rays. Times are median milliseconds; ρ is the median matched ratio. The pair count applies separately to each protocol. At $n = 32$ there are three completed new/new rounds per protocol and no completed arrangement/planar pair.

At $n = 32$, the baseline warmup reaches the 15-second cooperative wall-time allowance in each protocol. The driver does not repeat that capped arm. It retains both unsuccessful warmups and reports the complete new-arm times, with no speedup ratio. Every arm uses the same clock policy, checked once per 256 callback invocations; the allowance is cooperative rather than an exact preemptive timeout.

The retained work counters give a concrete mechanism for the growing difference. Table 2 records completed prepared calls. The maintained arrangement still has increasingly many equality candidates to inspect and nonextreme positive candidates to reject. Across these particular family instances, the planar diagram retains three full-dimensional minimum cells and three internal segments, and emits seven rays. These are measured instance counts; their finite pattern is not used as a proof of an asymptotic family formula.

11.3 Knot sources, controls, and timing noise

For each listed frozen source, the driver maximizes

$$(\mathbf{1}_{\dim P=2}, p, k, R, \text{support})$$

lexicographically among the declared audited raw-nullity-three candidates. This is a reproducible selection rule for the stated stratum, with no claim of typical-knot performance. The declared

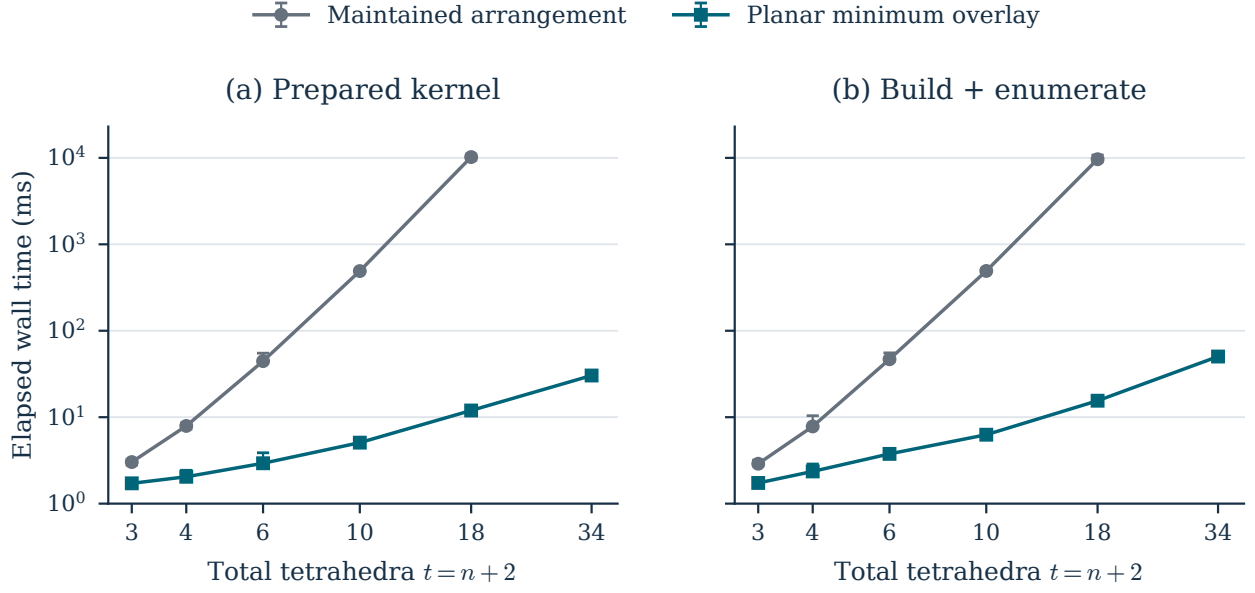


Figure 2: Observed complete-call medians for the principal family. Whiskers show the full range of measured primary-arm replicates, not confidence intervals. Both axes use logarithmic scaling. The arrangement curve stops at $n = 16$; its $n = 32$ warmup was capped and contributes no completed point. No asymptotic exponent is fitted.

n	Maintained arrangement				Planar overlay		
	Planes	Bases	Positive	Nonextreme	Cells	Segments	Rays
1	8	28	7	0	3	3	7
2	10	45	12	5	3	3	7
4	14	91	28	21	3	3	7
8	22	231	84	77	3	3	7
16	38	703	292	285	3	3	7

Table 2: Exact counters from complete calls. “Bases” means attempted arrangement bases; “Positive” includes candidates subsequently rejected by the standard-ray rank filter. The new output has no final rank-filter stage. Partial work at a cap is excluded from this table.

source `finite_trefoil` has no such candidate and is explicitly skipped; its interior-subdivision source does supply an eligible case. The three selected trefoil/figure-eight sources in Table 3 have zero essential-disc rays in their frozen complete reference lists. Thus they also supply negative disc-search controls at the supplied-sector level.

Source / control	t	R	Full median (ms)		Paired ρ	
			Old	New	Prep.	Full
Trefoil, interior subdivision	8	7	10.715	3.426	3.84	3.14
Figure-eight	10	7	17.835	4.176	5.73	4.18
Figure-eight, interior subdivision	13	7	100.276	11.717	12.13	9.30
Solid torus $\#\mathbb{RP}^3$	5	7	6.648	2.582	2.63	2.40
Solid torus $\#(S^2 \times S^1)$	7	7	21.271	5.503	4.64	3.87
<code>cap_1_2_3</code>	4	7	5.866	3.444	1.76	1.71
<code>interior_3_5</code>	6	2	7.531	1.515	6.42	4.90
Empty support	3	0	0.098	0.100	0.48	0.94
One allowed type	3	1	0.606	0.368	2.71	1.65

Table 3: Selected frozen sources and small-query controls. Every row has five complete pairs in each protocol and an independent exhaustive replay. “Full” includes kernel construction; “Prep.” uses a common prebuilt kernel. Exact source identifiers and supports are in the JSON.

All nine cap-type choices at $n = 2$ are included in the retained data; the $(1, 1)$ choice is already part of the principal family and is counted once. The empty-sector control was slower in this run: its prepared median times are approximately $1.42\ \mu\text{s}$ for the baseline and $3.56\ \mu\text{s}$ for the new call. At this scale individual durations are especially sensitive to measurement noise. Its full-call ratio below one is retained rather than discarded.

The 222 identical-planar (A/A) ratios have median 0.9967, minimum 0.1426, and maximum 4.6486. The two arms execute the same algorithm on the same input, so these extremes demonstrate timing variation, not algorithmic improvement. The minimum occurs in the empty-sector full control and the maximum in a small cap-type full control. Every outlier remains in the JSON. The sample sizes and this A/A spread support descriptive comparisons for these cases, with no narrow confidence interval or universal speedup claim.

11.4 Separate capability runs and independent-replay limits

The following larger principal-family cases receive one complete new-only build-and-enumerate call each, with a 90-second cooperative allowance. Their independent exhaustive replay has a separate 30-second allowance. These observations are neither paired timings nor speedup measurements, and no baseline is extrapolated.

n	t	Build (s)	Enumerate (s)	Total (s)	Independent replay
64	66	0.158	0.152	0.310	Verified (2.895 s)
128	130	0.954	0.409	1.364	Verified (16.990 s)
256	258	7.868	2.332	10.199	Capped (30 s); inconclusive

Table 4: Single new-only capability observations, all returning seven rays. The replay duration is outside the displayed total. The $n = 256$ producer finishes, but the independent complete-list replay reaches its separate cap; its coverage replay is therefore inconclusive.

At $n = 256$, all returned rays pass the producer’s native coordinate validation, and the exact producer theorem applies under its stated hypotheses. The timed experiment does not claim a completed independent coverage replay for that case. This distinction exposes a practical next target: after reducing enumeration work, dense preparation and independent verification can become the dominant costs.

The authoritative observations, source hashes, canonical ray digests, counters, arm orders, warmups, and limits are in `results/paired_benchmark.json`; the compact comparison table is `results/paired_benchmark.csv`. The reproduction driver is `experiments/benchmark.py`. Figures and this input fragment are regenerated by `experiments/make_article_assets.py`. Finite timing data are used to evaluate the implementation on the declared inputs; the asymptotic conclusions come from the proved bounds, and the global unknot-recognition coverage problem remains separate.

The exact arithmetic result is stronger than a timing observation: the new algorithm never needs the quartic all-equality candidate list, and the number of canonical outputs is bounded before the search runs. The timings determine whether that structural saving pays off on the tested implementation and inputs. Kernel preparation and independent replay are separate costs and may dominate once the local enumeration has become small.

12 What this changes for the quasi-polynomial program

12.1 A precise conditional composition

Theorem 12.1 (A low-nullity coverage implication). *Suppose an n -crossing knot diagram can be converted, with certified provenance, to a compact triangulated exterior of polynomial encoded size. Suppose further that a deterministic procedure constructs at most $2^{C(\log(n+2))^2}$ compatible sectors in that exterior, in that much time up to a polynomial factor, and that:*

1. *every constructed sector has matching nullity at most three;*
2. *if the diagram is an unknot, at least one constructed sector contains an essential standard vertex disc;*
3. *each emitted normal ray admits a polynomial-time exact essential-disc test in its binary encoding.*

Then complete search of these sectors decides unknottedness within $2^{O((\log(n+2))^2)} = n^{O(\log(n+2))}$ bit time.

Proof. Theorem 7.2, Corollary 5.5, and Lemma 7.3 give polynomial local enumeration and output encoding for each sector. The third hypothesis gives polynomial total testing per sector. Multiplying by the family-construction bound gives the stated complexity. An essential disc in a certified knot exterior certifies the unknot. The second hypothesis guarantees that exhausting the family without such a disc is a valid negative answer. $\square$

This is a transparent composition theorem, not a proof of its coverage hypothesis. The result isolates a remaining structural problem more sharply: the number of allowed quadrilateral coordinates can be large, yet the matching degrees of freedom of useful sectors may be small. One should seek

controlled sector discovery rather than infer that the disk itself must have logarithmically small quadrilateral support.

There is also a broader local observation for Euler-only queries. For $d \geq 1$, ordinary active-constraint enumeration tests at most $\binom{k}{d-1}$ systems for the corners of P . At a projective vertex, the active coordinate inequalities must span the $d - 1$ dimensional affine kernel directions; otherwise a small displacement in both signs would remain feasible. This argument also covers collapsed sections. Solving the candidate systems and applying Theorem 9.2 gives $k^{O(d)}$ local time with polynomial encoding factors. Thus logarithmic nullity, $d = O(\log(n + 2))$, and polynomial k already permit quasi-polynomial supplied-sector Euler queries. This is the standard active-constraint bound applied to the new objective reduction, not a new global recognition theorem. A potentially less restrictive coverage program would seek useful positive-Euler sectors of logarithmic nullity at each of polynomially many certified simplifying stages. It would still need a complete coverage theorem and controlled transformations; a positive Euler value alone cannot replace either requirement.

12.2 Why the missing hypothesis is substantial

The normal-sector family can still contain 3^t full type assignments. Nothing in a two-dimensional projective picture bounds how many distinct pictures must be constructed. One-vertex preprocessing removes cross-group interactions inside a low-nullity sector, but does not bound the nullity or the number of sectors. Sparse support is also not a universal remedy: the preceding project work gives layered-solid-torus examples with dense meridian support but very small supplied-sector nullity [8, 9].

Likewise, a compressed topological operation can run in polynomial encoded time while the number of hierarchy restarts remains large. Lackenby’s 2026 paper illustrates exactly why hierarchy length, pattern complexity, and intermediate encodings must be tracked separately [6]. This article supplies a complete local primitive with useful bounds; it supplies neither a new global restart potential nor a theorem that all necessary sectors are low-dimensional.

13 Further research questions and concrete next experiments

1. **Geometric realization of the quadratic interaction term.** Can rational strip systems attaining (9) be represented by normal matching equations of compact orientable three-manifolds with only $O(A)$ allowed quadrilateral types? Can this already happen in solid tori or certified knot exteriors? A positive construction would establish an intrinsic quadratic barrier; a negative theorem could improve the current bound.
2. **One-vertex low-nullity coverage.** For one-vertex triangulated knot exteriors, characterize the standard vertex discs whose occupied sector has nullity at most three. Does a controlled retriangulation always produce a useful such disc, or are there explicit families requiring arbitrarily large nullity? The test must distinguish support size from kernel dimension.
3. **Implement effective-dimension reduction.** Add the one-LP forced-zero-coordinate preprocessor and its rational primal/dual support witnesses from Corollary 6.4, followed by Proposition 6.3. Measure how many of the 188 excluded selected sectors become eligible. The benefit must include the preprocessing cost and rejected cases.
4. **A faster planar minimum constructor.** Replace all-cell halfplane clipping by a certified lower-hull construction and an exact overlay data structure. The output and skeleton bounds are already available. A successful implementation should improve the cubic arithmetic construction without weakening degeneracy handling or the independent interval checker.
5. **Reuse kernels across neighboring sectors.** Changing one allowed quadrilateral type modifies a bounded set of face labels. Can graph potentials and low-rank cycle systems be updated with a verifiable sparse certificate, while preserving a polynomial bound on coefficient growth? This directly targets the cost revealed when local enumeration is no longer dominant.
6. **Structure in four-dimensional kernels.** At nullity four, the normalized domain is a three-polytope and individual minimum complexes can already have more complicated output. Determine which interaction statistics replace (9). Prove bounds on intermediate faces as well as final rays before implementing an enumerator based only on final-output intuition.
7. **Euler-corner pruning and disc-directed traversal.** Implement Theorem 9.4 with a separate complete-corner negative certificate, and measure its cost on both rejected and surviving sectors. A positive corner need not be an essential disc. Can its topological outcome guide certified crushing or a lazy traversal of additional standard rays? Include knotted and no-disc sectors where positive-Euler candidates supply no direct witness.
8. **Integrate with certified hierarchy transitions.** Determine which compressed cut or retriangulation operations preserve low-nullity sectors, how their coordinate and peripheral data transport, and what potential charges the cost of rebuilding the planar models. The goal is an amortized global theorem, not another isolated local speedup.
9. **Formal proof of the small trusted geometry core.** The relevant core consists of affine interval intersection, closed convex clipping, the coordinate-face correspondence, and the planar crossing count. A Lean formalization could first verify the generic algebraic interfaces and certificate rejection conditions, before undertaking the larger normal-surface topology library.

A Reproduction and package layout

The archive contains this source and its compiled PDF, the new modules, focused tests, a pinned runnable source snapshot, an additive repository patch, exact fixtures, raw results, benchmark replicates, and scripts. The included README.md gives the shortest complete commands. The intended integration root is `Topology/UnknotRecognition/fast` at the pinned commit.

The essential source files are:

- `fastunknot/sector_planar.py`: exact producer and preprojected lift;
- `fastunknot/sector_planar_verify.py`: independent dense and minimum-tie-interval checking;
- `fastunknot/sector_planar_certificate.py`: enumeration certificates and supplied-sector essential-disc search;
- focused test modules, plus `experiments/audit.py`, `experiments/benchmark.py`, and exact family constructors.

The geometry and producer require only the Python standard library and the pinned native source. Regina is optional for fresh independent enumeration and is not imported by the runtime producer. Building the article requires a standard L^AT_EX installation with the packages in the preamble. Exact plotting scripts produce vector PDF figures; no generated raster image is used for a mathematical construction.

```
python3 reproduce.py --quick
python3 experiments/audit.py --fast code \
  --fresh-regina --output results/reproduced_audit.json
python3 experiments/benchmark.py --help
cd article
latexmk -pdf -interaction=nonstopmode -halt-on-error planar_sector_overlays.tex
```

The frozen data and scripts disclose their source hashes and random seeds. Checksums cover the delivered files; they are integrity records, not a substitute for rerunning the mathematical tests. The source checkout and baseline were not silently updated during the task. No remote repository mutation is part of this package.

B Limits of the experimental conclusions

The selected corpus is designed, finite, and inherited from a prior study; it is not a random sample of knots or triangulations. Several sources are solid tori or reducible controls, which are useful for observer tests but are not hard unknot-diagram inputs. A sector query has different costs from diagram conversion, sector selection, hierarchy repair, and complete recognition. Repeated timings of the same exact list control local implementation comparisons, not every possible workload.

The implementation uses exact rational arithmetic and checks source coordinates, but the article is an English mathematical proof, not a machine-checked formal proof. The new verifier is independent of the producer algorithm and chart; it intentionally shares the native triangulation parser and an older dense matching model. This boundary is recorded so future formalization or independent reimplementations can target it explicitly.

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- [9] ProveIt, *New deliveries and support-sensitive normal-disc discovery*, maintained synthesis chapter `incoming_sectors.tex` at commit `8188525b70033dcfe7c51ea5ae2c8723ad0c0198`. Includes the graph kernel, height bound, dense-sector search, and sparse-support obstruction. [Maintained research tree at the pinned commit](#).